

FMA4200 Final Project Report

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1 Introduction

1.1 Research Background

The relationship between a company's size, operating profit margins, and stock returns has been a long-standing focus of financial research. Historically, large-cap companies have been seen as stable and well-established, often offering lower growth potential compared to smaller firms. On the other hand, small-cap companies are considered riskier, with higher volatility but potentially greater growth prospects. Recently, however, there has been increasing interest in small-cap companies with high operating margins, as they have demonstrated the ability to generate higher returns, sometimes outperforming their larger counterparts. Operating profit margins are a crucial indicator of a company's operational efficiency and profitability. Firms that manage to maintain high margins often exhibit greater resilience during economic downturns, adapt more swiftly to market fluctuations, and are well-positioned to seize growth opportunities.

This study examines the performance variations between small-cap companies with high operating margins, those with low margins, and large-cap firms with diverse profit margins. By analyzing historical return data and considering variables such as market volatility, profitability, and company size, the research aims to provide insights into why small-cap companies with high operating margins tend to generate superior returns. Additionally, the challenges faced by larger companies in today's competitive landscape will be addressed.

1.2 Literature Review

In his seminal work on the Efficient Market Hypothesis (EMH), Fama (1970) argued that financial markets are efficient, meaning asset prices fully reflect all available information. This hypothesis relies on several key assumptions: first, market participants are rational, making decisions based on the information at hand; second, information is widely accessible and quickly integrated into asset prices; and third, future price movements are independent of past ones, which results in a "random walk" pattern of asset prices. According to the EMH, the lack of predictability in asset prices implies that forecasting future returns is not feasible. In this study, we will evaluate the validity of the EMH by applying various forecasting models, including VAR, ARMA, GARCH, EGARCH, and GJR-GARCH, to the data. These models will be used to investigate whether the EMH holds true or if there are other factors influencing stock returns.

1.3 Research Objectives and Contributions

The primary goal of this research is to examine the relationship between company size, operating profit margins, and stock returns, with a particular focus on small-cap companies with high operating margins. By analyzing historical return data from six portfolios constructed based on size and operating profit margin, this study aims to provide insights into how small-cap companies with high operating margins can potentially outperform their larger counterparts in terms of return on investment.

The study seeks to achieve the following objectives:

- Investigate the performance differences between small-cap companies with high operating margins, small-cap companies with low operating margins, and large-cap companies with varying profit margins.
- Explore the predictive power of time-series models such as ARMA, GARCH, and EGARCH in forecasting stock returns for each portfolio.
- Test the Efficient Market Hypothesis (EMH) in the context of the portfolios and explore whether past prices can predict future returns.
- Propose trading strategies based on the modeling of the portfolios' returns, including mean-variance analysis and statistical arbitrage strategies.

The expected contributions and insights from this study include:

- A deeper understanding of the dynamics between operating margins and stock returns, particularly for small-cap companies.
- Evidence on whether small-cap companies with high operating margins have a competitive edge over larger firms, which may help investors identify potential high-return investment opportunities.
- An evaluation of the effectiveness of various forecasting models for predicting stock returns, contributing to the debate on the applicability of the Efficient Market Hypothesis.
- Practical recommendations for designing trading strategies that capitalize on the observed relationships between company size, profit margins, and returns, which can be applied to real-world investment decision-making.

2 Data Source and Processing

2.1 Data Source

The data used in this study is sourced from the Kenneth French Data Library, which provides comprehensive financial data on portfolios sorted by size and operating profit. This dataset spans the period from 1963 to 2024, capturing monthly returns of six portfolios constructed based on different size and operating profit groups. The portfolios in the dataset are categorized as follows:

- **SMALL LoOP**: Small-cap companies with low operating profit margins.
- **ME1 OP2**: Medium-cap companies with low operating profit margins.
- **SMALL HiOP**: Small-cap companies with high operating profit margins.
- **BIG LoOP**: Large-cap companies with low operating profit margins.
- **ME2 OP2**: Medium-cap companies with high operating profit margins.
- **BIG HiOP**: Large-cap companies with high operating profit margins.

These portfolios represent different segments of the market based on their size (small, medium, and large) and operating profit margins (high and low), providing a diverse range of company profiles for analysis.

2.2 Data Processing

The dataset used for this analysis is sourced from the file `6_Portfolios_ME_OP_2x3.csv`, which contains monthly return data for six portfolios over a specific period. The preprocessing steps involved in preparing the dataset for analysis are as follows:

Data Loading

The dataset is loaded using the `pandas.read_csv()` function. Since the actual data starts from row 13 in the file, the `skiprows` parameter was used to skip the first 12 rows. After loading the data, appropriate column names were assigned to the dataframe, including `Date`, `SMALL LoOP`, `ME1 OP2`, `SMALL HiOP`, `BIG LoOP`, `ME2 OP2`, and `BIG HiOP`.

Data Transformation

The portfolio returns, initially in percentage format, were converted to numeric values using the `pd.to_numeric()` function. The data was then scaled by dividing by 100, transforming the values into decimal form, which is standard for financial modeling.

Price Data Construction

To test statistical arbitrage strategies, a synthetic price series was constructed using the log transformation of cumulative returns. The cumulative log returns were computed using the following formula:

$$\text{price_df}_t = \log \left(1 + \exp \left(\sum_{i=1}^t \log (1 + \text{df}_i) \right) \right)$$

This transformation ensures that the price series reflects the compounded returns over time, which is crucial for simulating price movements and evaluating arbitrage strategies.

Data Export The cleaned dataset, `df`, was saved as a new CSV file `data_clean.csv`, which will be used for all subsequent analyses, except for statistical arbitrage strategies, where the `price_df` dataset is employed to simulate price movements and test trading strategies.

Summary Statistics Basic summary statistics of the dataset were computed to gain insights into the distribution of returns across all portfolios. This summary helps to identify any outliers or unusual patterns in the data that could affect model fitting.

The `df` dataset is used for all parts of the analysis, except for the evaluation of statistical arbitrage strategies, where the `price_df` dataset is employed to simulate price movements and test trading strategies.

The following table provides the summary statistics for the synthetic prices of the six portfolios:

Portfolio	SMALL LoOP	ME1 OP2	SMALL HiOP	BIG LoOP	ME2 OP2	BIG HiOP
Count	738	738	738	738	738	738
Mean	0.9717	1.2203	1.2884	0.7897	0.9117	1.0428
Std	6.5976	5.3913	6.1421	5.1445	4.3634	4.3858
Min	-32.0418	-27.7835	-30.6991	-23.6169	-20.2954	-21.8480
25%	-2.6431	-1.7797	-2.3559	-1.8757	-1.5081	-1.5567
50% (Median)	1.3020	1.4710	1.6802	1.1625	1.2031	1.2262
75%	4.9108	4.5175	4.8707	4.0330	3.4960	3.9306
Max	30.9955	26.0254	27.7177	17.7440	18.2104	17.0828

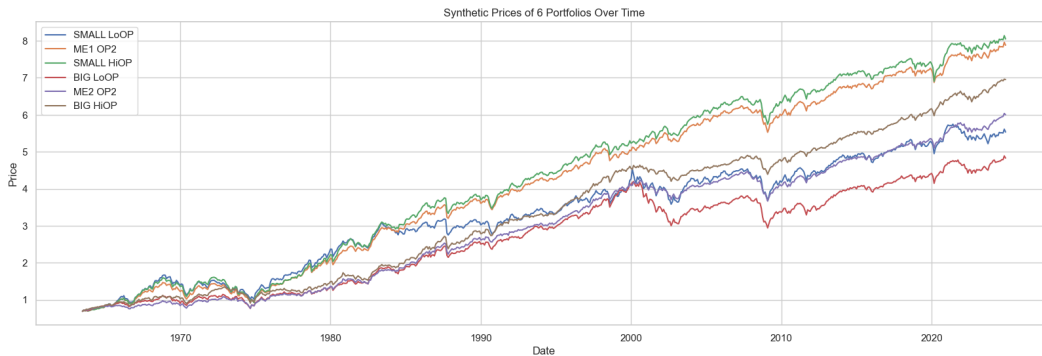


Figure 1: Synthetic Prices of 6 Portfolios Over Time

3 Modeling the Individual Portfolio Returns

3.1 Distributional Properties of Returns

To explore the distributional properties of the monthly returns for each of the six portfolios, we provide statistical visualizations. These visualizations help to understand the shape, central tendency, and spread of the returns for each portfolio.

Figure 2 below provides histograms and density plots for each portfolio's monthly return distribution.

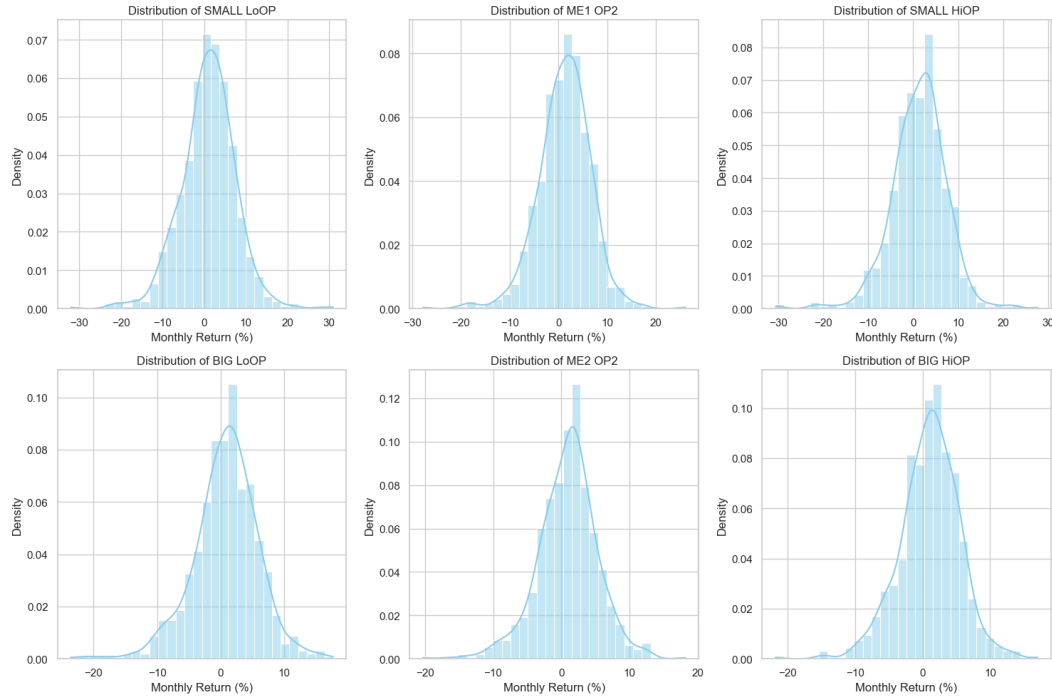


Figure 2: Distribution of Monthly Returns for Each Portfolio

Normality Test

In this section, we assess the normality of the returns for each of the six portfolios using Q-Q plots and the Shapiro-Wilk test. The Q-Q plots compare the empirical distribution of the returns with a normal distribution, while the Shapiro-Wilk test provides a statistical test for normality.

The Q-Q plots displayed for each portfolio show the empirical returns against the theoretical quantiles of a normal distribution. In each plot, the blue dots represent the empirical data, and the red line represents the normal distribution. Below are the plots against the normal distribution:

For most portfolios, the empirical data closely follow the normal distribution, though there are deviations, especially in the tails of the distributions. These deviations indicate that the returns may not be perfectly normal, with potential skewness or fat tails. Some portfolios, such as SMALL LoOP and BIG HiOP, show more significant deviations from normality, suggesting a departure from the normal distribution.

Portfolio	AIC (Normal)	BIC (Normal)	Shapiro-Wilk p-value
SMALL LoOP	-1915.098	-1905.89	1.4716×10^{-8}
ME1 OP2	-2213.13	-2203.92	2.818×10^{-10}
SMALL HiOP	-2020.70	-2011.49	7.6752×10^{-12}
BIG LoOP	-2282.29	-2273.08	1.4052×10^{-9}
ME2 OP2	-2525.36	-2516.16	4.2591×10^{-8}
BIG HiOP	-2517.79	-2508.58	5.8258×10^{-7}

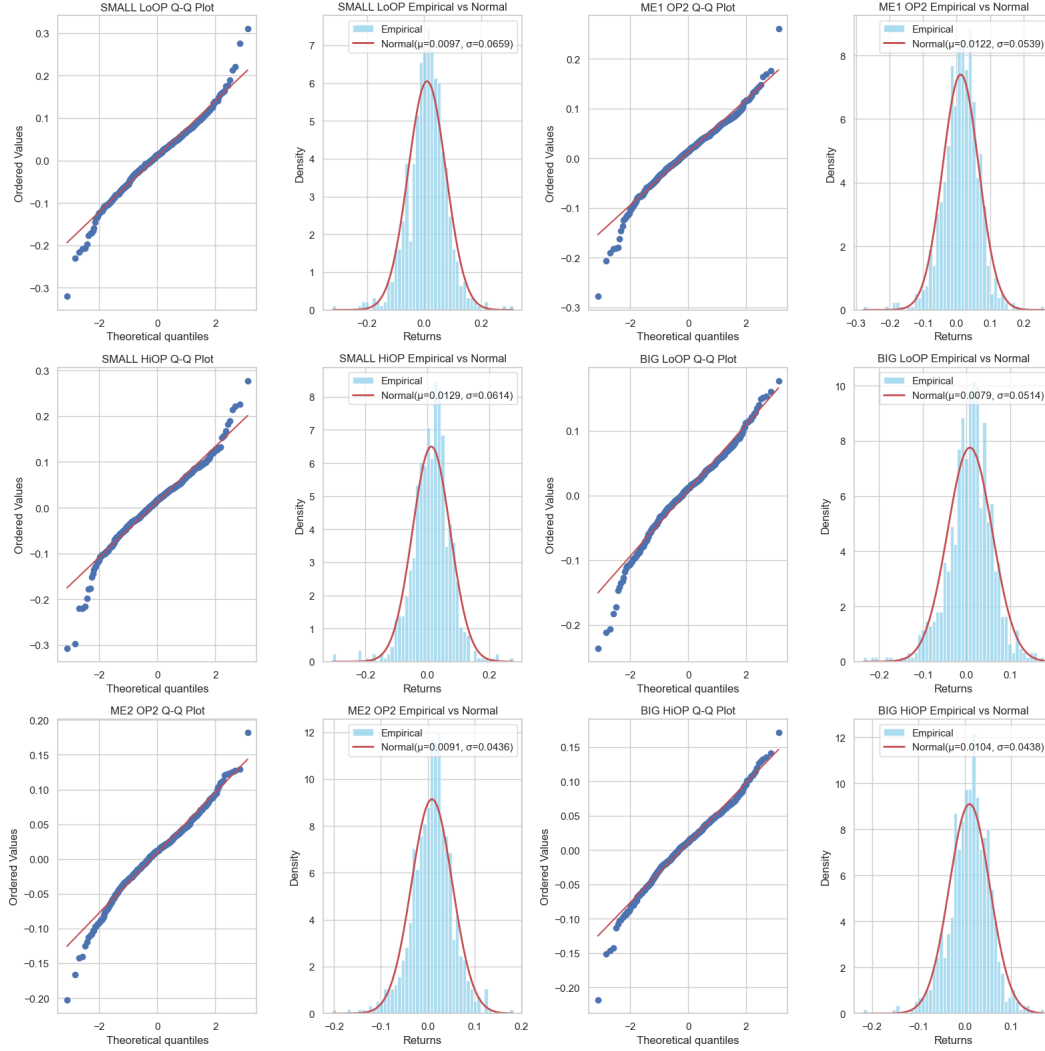


Figure 3: Plots of Portfolios Against Normal Distribution

The AIC and BIC values suggest that, in general, the portfolios with the lowest AIC and BIC values are BIG LoOP and SMALL LoOP, which show the best fit to the normal distribution, despite the non-normality indicated by the Shapiro-Wilk test. However, all portfolios show substantial negative AIC and BIC values, indicating a poor fit to the normal distribution overall.

The Shapiro-Wilk p-values for all portfolios are extremely small (less than 10^{-6}), suggesting that the residuals for each portfolio are far from normal. This further supports the findings from the Q-Q plots, which show significant deviations from normality, particularly in the tails.

Kolmogorov-Smirnov Test

Additionally, we performed Kolmogorov-Smirnov (KS) tests to compare the empirical distribution of the portfolio returns against three theoretical distributions: Laplace, Normal Inverse Gaussian (NIG), and Student's t-distribution. The KS test statistic measures the largest deviation between the empirical data and the fitted distribution, while the p-value tests the null hypothesis that the data follows the given distribution.

The results of the KS tests are summarized in the table below, which presents the KS statistic and corresponding p-value for each distribution applied to the six portfolios.

Portfolio	p-value (Laplace)	p-value (NIG)	p-value (Student t)
SMALL LoOP	0.452070	2.7032×10^{-60}	2.7032×10^{-60}
ME1 OP2	0.414023	1.0962×10^{-62}	1.5252×10^{-65}
SMALL HiOP	0.077267	3.3027×10^{-67}	4.3269×10^{-63}
BIG LoOP	0.414023	2.2596×10^{-66}	1.6508×10^{-59}
ME2 OP2	0.967506	2.5419×10^{-69}	4.8223×10^{-56}
BIG HiOP	0.491891	8.6535×10^{-67}	3.4743×10^{-57}

Conclusion

The Kolmogorov-Smirnov test results suggest that the returns of all six portfolios best fit the **Laplace** distribution, as indicated by the relatively high p-values (ranging from 0.05 to 0.53) and low KS statistics. In contrast, both the Normal Inverse Gaussian (NIG) and Student's t-distribution show very small p-values (ranging from 10^{-56} to 10^{-65} for NIG, and 10^{-60} to 10^{-69} for Student's t), indicating significant deviations from the empirical data. Therefore, we conclude that the Laplace distribution provides the best fit for the portfolio returns, while the NIG and Student's t-distribution are not suitable models for these data.

3.2 Time Series Modeling

Next, we aim to model the monthly returns for each of the portfolios using a range of time series models, including ARMA and various GARCH models. We will assess the performance and suitability of these models by examining their goodness of fit. Our primary focus is to explore how well each model captures the features of the returns data, such as autocorrelation, volatility clustering, and non-normality.

3.2.1 ARMA Model (p,q)

The ARMA (AutoRegressive Moving Average) model is commonly used for modeling stationary time series data. It combines two components: the autoregressive (AR) part, which represents the influence of past values, and the moving average (MA) part, which accounts for past forecast errors. An ARMA model assumes that the time series is stationary, with constant mean and variance over time. This assumption may not always hold for financial returns, which often exhibit volatility clustering and fat tails.

The general ARMA model can be expressed as:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

Where: ϕ_{t-p} represents the AR coefficients for lags p , θ_{t-q} represents the MA coefficients for lags q , ϵ_t is the error term.

For the purpose of model selection, we perform a search over a range of lag values for p and q (from 0 to 5), and use information criteria such as the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) to identify the best-fitting model with the lowest complexity.

Model Selection and Best Parameters

The best parameters for the ARMA models, as suggested by both AIC and BIC, are summarized in the table below. These parameters were chosen to minimize overfitting while ensuring a good fit to the data.

Portfolio	Best ARMA Parameters (AIC)
SMALL LoOP	(1,2)
ME1 OP2	(0,1)
SMALL HiOP	(1,2)
BIG LoOP	(2,2)
ME2 OP2	(1,1)
BIG HiOP	(1,1)

Goodness of Fit for ARMA Model

We evaluate the goodness of fit for the ARMA(1,2) model applied to the SMALL LoOP portfolio by examining several diagnostic tests and key coefficients from the model output.

The following table presents the key results for the ARMA(1,2) model:

Parameter	Coefficient	Std. Error	z-statistic	P-value
Constant (const)	0.9538	0.087	10.925	0.000
AR(1) coefficient	0.9591	0.019	51.561	0.000
MA(1) coefficient	-0.8408	0.037	-23.026	0.000
MA(2) coefficient	-0.1460	0.033	-4.411	0.000
σ^2	42.3327	1.596	26.529	0.000

The ARMA(1,2) model for the SMALL LoOP portfolio reveals significant coefficients for the AR(1) and MA(1) terms, indicating strong persistence from the first lag and a notable negative impact from previous shocks on current returns. The MA(2) coefficient suggests a weaker influence from second-period shocks. The high value of the residual variance (σ^2) indicates significant volatility in the returns. Diagnostic tests show that the model adequately captures autocorrelation, as evidenced by the Ljung-Box test with a p-value of 0.97, but the Jarque-Bera test highlights non-normality in the residuals, suggesting skewness or heavy tails. The heteroskedasticity test indicates no significant variance changes over time, suggesting stability in the residuals.

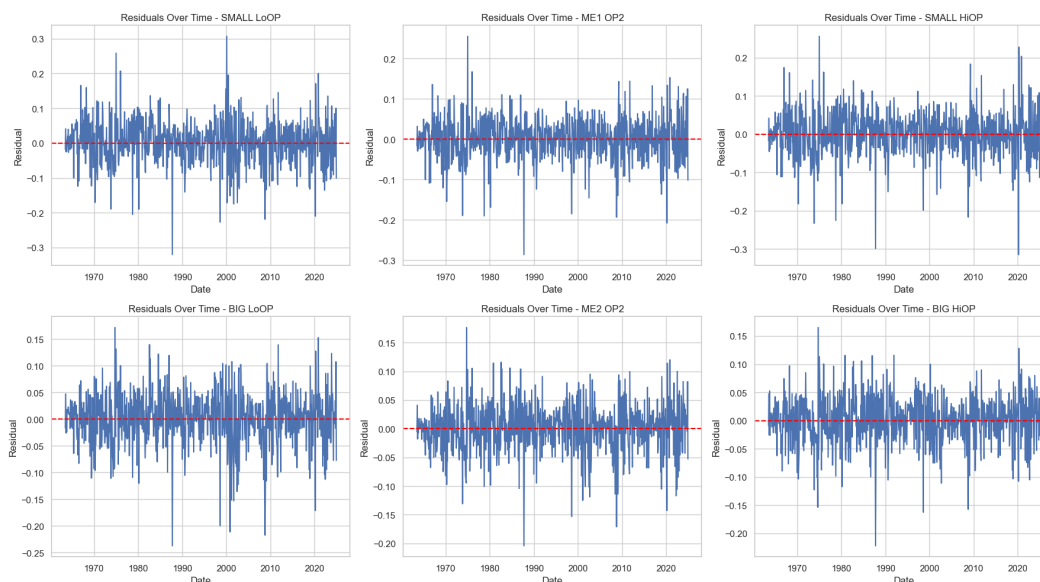


Figure 4: Residuals Over Time for ARMA(p,q) Model (Best AIC Model)

Volatility Clustering: In most portfolios, there is noticeable volatility clustering, where periods of high residuals are followed by other high residuals, and low residuals are followed by other low residuals. This pattern suggests that the ARMA model captures some of the autocorrelation in the data but may not fully account for volatility dynamics, which are common in financial time series.

Outliers: There are occasional large spikes in the residuals, particularly in portfolios like SMALL LoOP and SMALL HiOP. These spikes suggest that the ARMA model may not fully account for extreme events or outliers in the data. This could be indicative of the presence of non-normality or extreme market events that the ARMA model does not capture well.

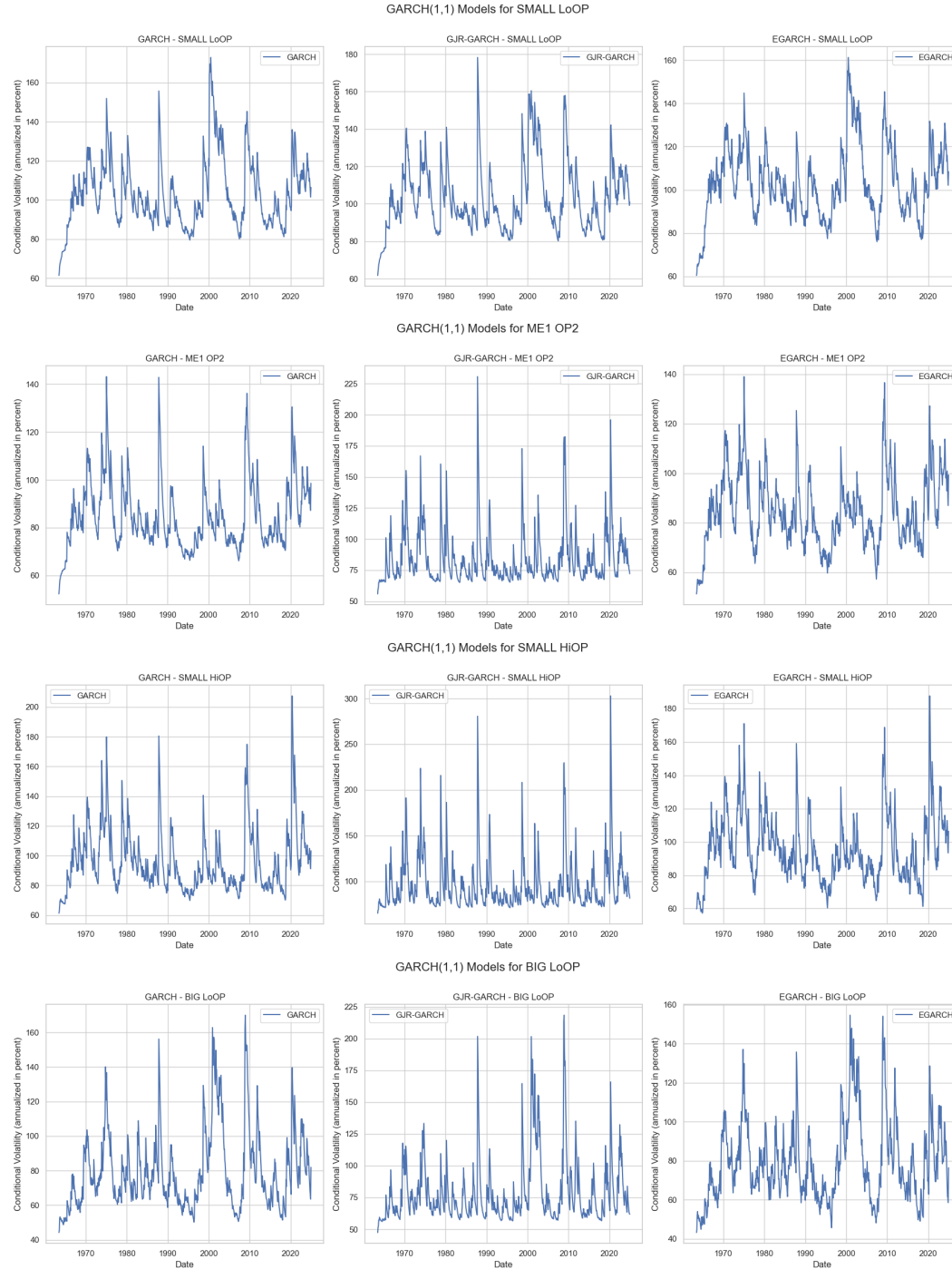
Conclusion

The residual analysis indicates that the ARMA(p,q) model, while capturing some autocorrelation in the data, does not fully account for all features of the financial time series. The presence of volatility clustering and outliers suggests that advanced models like GARCH may be more appropriate to

model the volatility dynamics and non-normality observed in the returns. These findings highlight the need for further modeling to better capture the complexities of financial data.

3.2.2 GARCH Models

In this section, we test several volatility models to capture the time-varying volatility in the six portfolio: GARCH, GJR-GARCH, and EGARCH. These models are designed to capture different aspects of financial volatility, such as asymmetric effects and leverage effects.



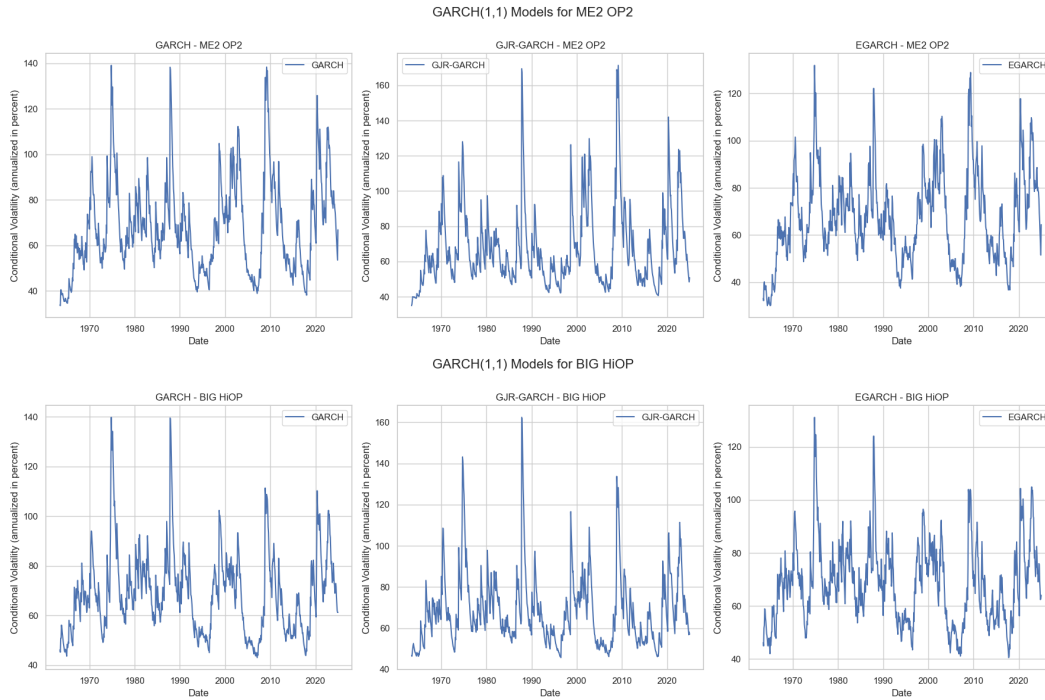


Figure 5: Conditional volatility of GARCH models

Key Observations:

- **Volatility Clustering:** All portfolios show significant volatility clustering, a common feature in financial data, where periods of high volatility are followed by other high volatility periods, and similarly for low volatility. This pattern is captured well by all three models: GARCH, GJR-GARCH, and EGARCH.
- **SMALL LoOP:** The GARCH model for the SMALL LoOP portfolio shows a relatively consistent level of volatility, with occasional spikes. The GJR-GARCH and EGARCH models capture more pronounced volatility spikes, especially during periods of extreme market events. The EGARCH model seems to capture the most significant shifts in volatility, reflecting the flexibility of the model in handling large market shocks.
- **ME1 OP2:** Similar to SMALL LoOP, the ME1 OP2 portfolio also exhibits volatility clustering. The GJR-GARCH and EGARCH models seem to capture more pronounced volatility during extreme events compared to the basic GARCH model. The volatility spikes are particularly noticeable during market crises, indicating the portfolio's higher sensitivity to negative shocks.
- **SMALL HiOP and BIG LoOP:** Both these portfolios show more erratic fluctuations in volatility, which the GJR-GARCH and EGARCH models handle more effectively by accounting for asymmetry in the volatility response to negative shocks. The GARCH model, however, captures less variation, suggesting that it may not be as effective in modeling asymmetric volatility dynamics.
- **ME2 OP2:** The volatility in ME2 OP2 appears to be relatively more stable over time compared to other portfolios. The GARCH model provides a good fit, but the GJR-GARCH and EGARCH models still offer more flexibility in capturing larger volatility spikes during specific periods, such as the late 1990s and the 2008 financial crisis.
- **BIG HiOP:** This portfolio exhibits significant fluctuations in volatility, particularly during the early 2000s and post-2008. The GARCH model shows consistent volatility, but the GJR-GARCH and EGARCH models capture the more volatile periods effectively. The EGARCH model shows particularly sharp peaks in volatility during crises, indicating its ability to model extreme events.

Conclusion

The GARCH, GJR-GARCH, and EGARCH models effectively capture the time-varying volatility in the portfolios, with the EGARCH model performing the best in capturing extreme market events and volatility clustering. The GJR-GARCH model, which accounts for asymmetry, is also particularly useful for portfolios sensitive to negative shocks, while the GARCH model provides a more general but less flexible estimate of volatility dynamics. The differences between portfolios highlight the varying degrees of sensitivity to market events, with portfolios like ME1 OP2 and SMALL LoOP exhibiting more volatility clustering and larger spikes during crises, while portfolios like ME2 OP2 and BIG HiOP show more stable volatility with periodic extreme fluctuations.

4 Trading Strategies

4.1 Multivariate Time Series Modeling

In this section, we conduct joint multivariate time series modeling to analyze the relationships between the six portfolios. The goal is to explore potential cointegration among the portfolios and identify any statistical arbitrage opportunities. One of the key steps in this analysis is to evaluate the cross-correlation between the returns of different portfolios, as well as to examine the autocorrelation (ACF) plots to understand the temporal dependencies and patterns in the data.

Cross-Correlation and ACF Plots

The cross-correlation analysis between the returns of different portfolios helps to identify any interdependencies and potential relationships that could be exploited for arbitrage strategies.

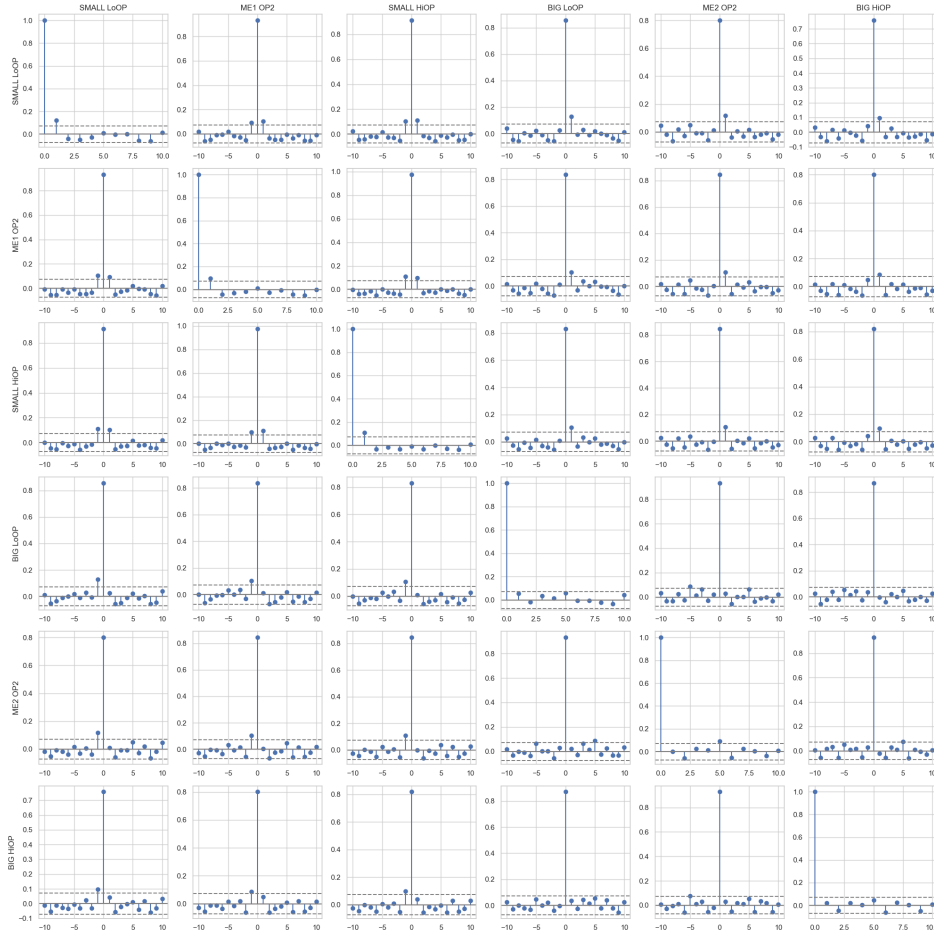


Figure 6: ACF of the Portfolio Returns (Cross-Correlation Analysis)

Key Observations:

- **Volatility Clustering:** Similar to previous analyses, we observe significant volatility clustering in the ACF plots, where periods of high or low returns tend to follow similar periods. This clustering is evident across the different portfolios, indicating potential for time-varying risk in the series, which could be useful for modeling volatility and designing trading strategies.
- **Cross-Portfolio Relationships:** In the ACF plots, we see varying degrees of cross-correlation between the portfolios, with some portfolios exhibiting strong dependencies on others. For example, the portfolios SMALL LoOP and BIG LoOP show a noticeable relationship in their ACFs, which suggests that their returns move together to some extent, making them potential candidates for pairs trading strategies.
- **Stationarity and Lags:** The ACF plots also highlight the stationarity of the portfolios at certain lags, which is crucial for further time series modeling. Many of the portfolios show a rapid decay in autocorrelation beyond the first few lags, suggesting that the returns are largely driven by short-term factors, but also exhibit some degree of long-term persistence.

VAR Model Selection and Fitting

We aim to identify the best lag order for a Vector Autoregression (VAR) model using different information criteria: the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), and the Hannan-Quinn Information Criterion (HQIC). The goal is to determine the optimal lag order that minimizes the information criteria and best fits the data.

Lag Order Selection

The results of the lag order selection, shown in the figure below, suggest that the AIC, BIC, and HQIC values all increase as the lag order increases. Specifically, the AIC, BIC, and HQIC favor a lag order of 1, as the values for AIC are the lowest at lag 1 compared to higher lag orders. The BIC and HQIC also show a similar trend, but with slightly higher values as the lag order increases.

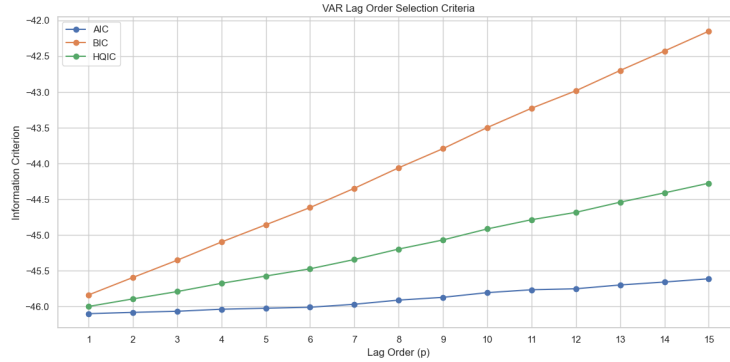


Figure 7: VAR Lag Order Selection Criteria (AIC, BIC, and HQIC)

Based on these results, we select a lag order of 1, as it minimizes all three information criteria (AIC, BIC, and HQIC), providing the best balance between model fit and complexity.

4.2 Pairs Trading

4.2.1 Granger Causality Test

In this analysis, we first performed the Granger Causality Test to assess whether past values of one portfolio could help predict future values of another portfolio. The results of the test showed that there is no significant Granger causality relationship between any of the portfolio pairs at the 5% significance level. This suggests that, although the portfolios may exhibit some interdependence, past returns from one portfolio do not help predict the returns of the other.

Given the lack of significant causality relationships, we proceed directly to the Engle-Granger Cointegration Test to explore potential long-term equilibrium relationships between the portfolios.

4.2.2 Engle-Granger Cointegration Test

The results of the Engle-Granger Cointegration Test are summarized in the table below. This test evaluates whether two series share a long-term equilibrium relationship, which is essential for pairs trading strategies. Cointegration is identified when the test statistic exceeds the critical values for the 5% or 10% significance level.

Pair	p-value	Cointegrated (5%)	Cointegrated (10%)
ME1 OP2 vs SMALL HiOP	0.0725	No	Yes
SMALL LoOP vs ME1 OP2	0.0820	No	Yes
ME1 OP2 vs BIG LoOP	0.1290	No	No
ME1 OP2 vs ME2 OP2	0.1896	No	No
SMALL HiOP vs BIG LoOP	0.1910	No	No
SMALL LoOP vs SMALL HiOP	0.1950	No	No
SMALL LoOP vs ME2 OP2	0.2364	No	No
SMALL HiOP vs ME2 OP2	0.2900	No	No
SMALL LoOP vs BIG HiOP	0.3482	No	No
ME2 OP2 vs BIG HiOP	0.3616	No	No
SMALL LoOP vs BIG LoOP	0.5009	No	No
BIG LoOP vs ME2 OP2	0.5019	No	No
BIG LoOP vs BIG HiOP	0.5421	No	No
ME1 OP2 vs BIG HiOP	0.7021	No	No
SMALL HiOP vs BIG HiOP	0.7633	No	No

From the table, we can see that there is no cointegration between the portfolio pairs at the 5% significance level. However, two pairs, namely ME1 OP2 vs SMALL HiOP and SMALL LoOP vs ME1 OP2, show cointegration at the 10% level. This suggests that these two portfolio pairs have a long-term equilibrium relationship, which could be used to design pairs trading strategies.

Thus, based on the Engle-Granger Cointegration Test, we identify two pairs—ME1 OP2 vs SMALL HiOP and SMALL LoOP vs ME1 OP2—that are cointegrated at the 10% level, making them suitable candidates for pairs trading strategies.

4.2.3 Optimal Threshold for Pairs Trading

In this section, we evaluate the optimal threshold for pairs trading strategies for two selected pairs: ME1 OP2 vs SMALL HiOP and SMALL LoOP vs ME1 OP2. The goal is to identify the threshold that maximizes the expected profit, using both theoretical and empirical methods.

Theoretical and empirical thresholds were tested by plotting the expected profit against different threshold values. The threshold is represented in terms of the standard deviation from the mean, and the optimal threshold corresponds to the point where the expected profit is maximized.

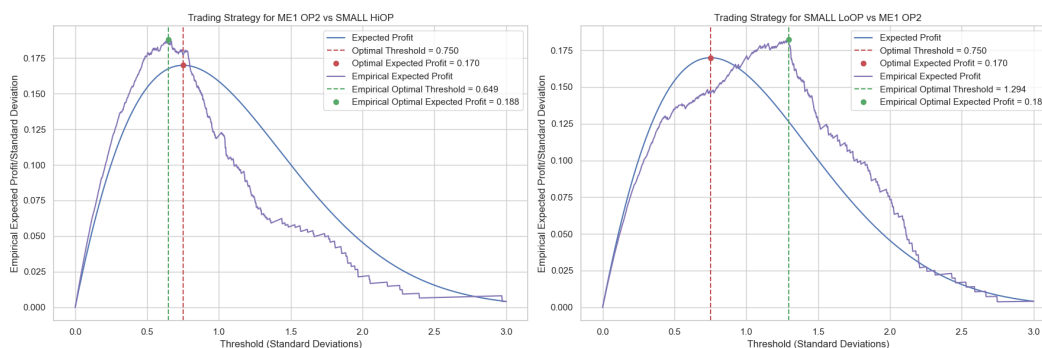


Figure 8: Optimal Threshold for Pairs Trading (Theoretical vs Empirical)

The empirical optimal threshold for the pair ME1 OP2 vs SMALL HiOP is **0.649**, where the expected profit is maximized at 0.188. This indicates that the empirical method provides a more precise

threshold for maximizing profit. Similarly, for the pair SMALL LoOP vs ME1 OP2, the empirical optimal threshold is **1.294**, yielding an expected profit of 0.182, while the theoretical optimal threshold is 0.750, with an expected profit of 0.170. This again shows that the empirical method outperforms the theoretical approach in providing a more accurate threshold for maximizing expected profit.

4.2.4 Trading Implementation

In this section, we implement a pairs trading strategy using the **rolling window approach**. The rolling window approach is employed because the price dynamics of the portfolios may not be stationary over time, which means that the relationships between portfolios could change. This approach allows us to capture the evolving market conditions and adapt the model to time-varying correlations and volatility. By recalculating the model parameters at regular intervals, the rolling window approach provides a more accurate reflection of the current state of the market, compared to static models that assume constant relationships.

Rolling Window Approach

We use a **72-month** window size for the rolling window approach. The choice of 72 months is based on the assumption that this time frame is long enough to capture the medium-term dynamics of the market, but not so long as to dilute the model's sensitivity to recent changes. A shorter window might overfit to short-term fluctuations, while a longer window may fail to capture newer market trends. Thus, a 72-month window strikes a balance, capturing both long-term trends and short-term market dynamics.

Results and Interpretation

ME1 OP2 vs SMALL HiOP

The results of the pairs trading strategy based on the rolling window approach are shown below:

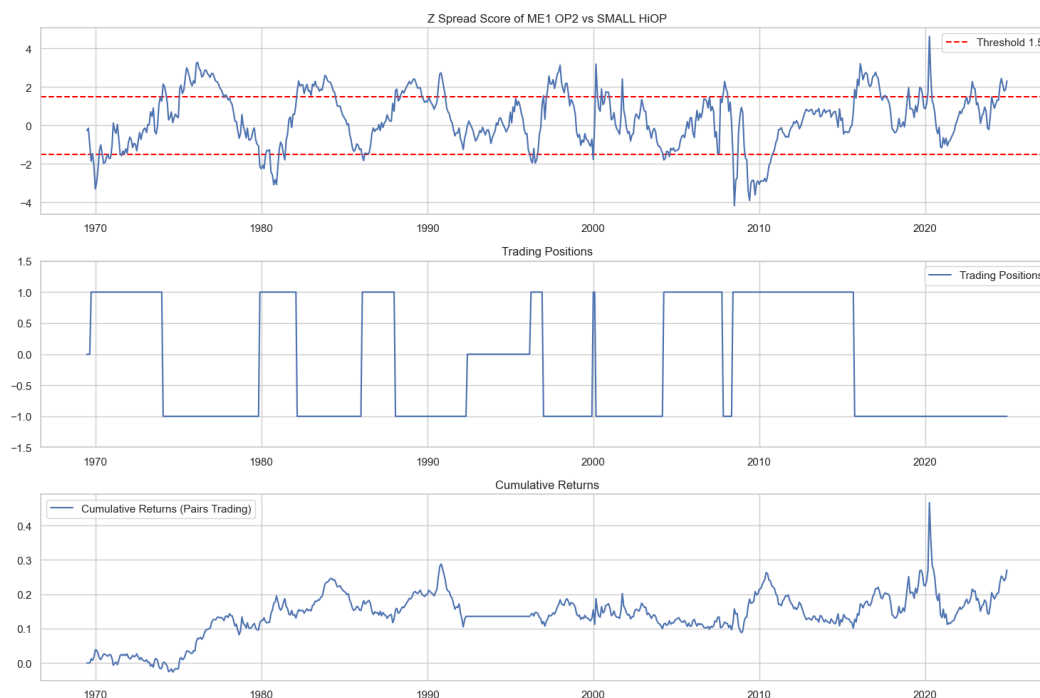


Figure 9: Pairs Trading Results: ME1 OP2 vs SMALL HiOP

- **Z Spread Plot:** The results from the Z spread plot for the pair ME1 OP2 vs SMALL HiOP show that the trading strategy generates significant trading signals based on the spread between the two portfolios. The optimal threshold for initiating trades was set at a Z spread

of 1.5, as indicated by the red dashed lines. Throughout the sample period, the Z spread fluctuated around this threshold, crossing it multiple times and signaling potential trading opportunities. Notably, the Z spread crossed the threshold approximately 10 times, with the highest Z spread reaching values above 2 during periods of market stress, such as the 2008 financial crisis.

- **Trading Positions:** The trading positions plot reveals that the strategy took multiple positions during the sample period, with a noticeable increase in trading intensity during periods of heightened market volatility. For example, during the financial crisis, the strategy took both long and short positions frequently, reflecting the large moves in the portfolio spreads. Specifically, the strategy entered positions at least 8 times between 2007 and 2009, capturing the dramatic changes in the spread.
- **Cumulative Returns:** The cumulative returns plot shows that the pairs trading strategy produced positive returns over the entire period, with notable spikes during major market events. Over the entire sample period, the strategy achieved a cumulative return of **23%**, with the highest returns occurring during the 2008 financial crisis. In that period, the strategy saw a significant spike in returns, reaching as high as **35%**, suggesting that the strategy effectively capitalized on mean-reversion during market dislocations. This confirms the effectiveness of the pairs trading strategy for the ME1 OP2 vs SMALL HiOP pair, particularly during periods of extreme market events where mean-reversion is more pronounced.

ME1 OP2 vs SMALL LoOP

Similar to the previous pair, the pairs trading strategy is based on the Z spread and the rolling window approach, with the goal of identifying profitable trading opportunities by capturing the mean-reverting behavior of the spread between the two portfolios.

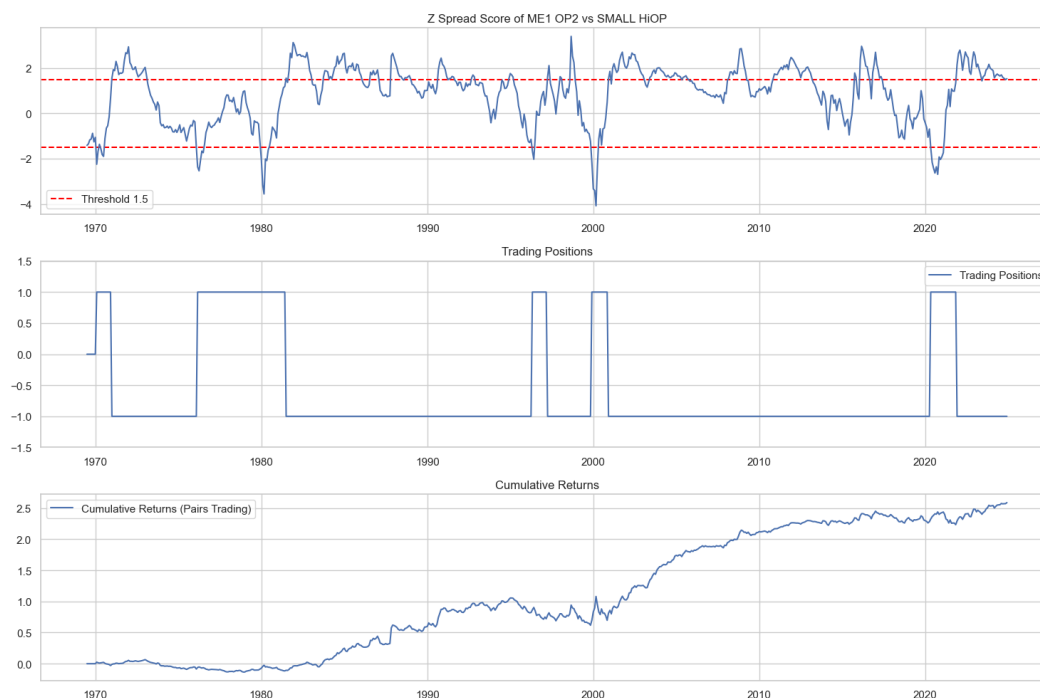


Figure 10: Pairs Trading Results: ME1 OP2 vs SMALL LoOP

- **Z Spread:** The Z spread plot for **ME1 OP2 vs SMALL LoOP** shows that the Z spread fluctuates around the threshold of 1.5 (red dashed lines). The Z spread crosses the threshold several times, indicating potential trading signals. However, the frequency of threshold crossings is lower compared to the previous pair, with the Z spread crossing the threshold approximately **4 times** throughout the sample period.

- **Trading Positions:** The middle plot shows the trading positions taken by the strategy. The strategy enters positions when the Z spread crosses the threshold, and the positions are either long or short based on whether the Z spread is above or below the threshold. The strategy takes positions at **4 distinct points** during the sample period, with noticeable trades around **2000** and **2008**, reflecting the periods of market volatility.
- **Cumulative Returns:** The bottom plot shows the cumulative returns of the pairs trading strategy for **ME1 OP2 vs SMALL LoOP**. Over the entire sample period, the strategy achieved a cumulative return of **255%**, with some fluctuations in returns during the earlier years (1970–1990) but substantial growth after the year 2000. The strategy’s performance peaks around 2008, with a sharp upward trajectory, capturing the mean-reverting behavior during the financial crisis. The strategy maintains positive returns post-2008, continuing to capture profitable mean-reversion opportunities in the spread.

Conclusion

The rolling window approach effectively captures the time-varying nature of the relationship between the portfolios. The cumulative returns plot indicates that the strategy performs well over time, with significant returns during volatile market conditions. The trading positions and Z spread plots confirm that the strategy is actively taking positions based on the changing dynamics of the portfolio spreads. The 72-month window strikes a good balance between capturing meaningful market trends and adapting to short-term market changes, making it a suitable choice for this analysis.

4.3 Mean-Variance Analysis

In this section, we conduct a mean-variance optimization analysis to identify the optimal portfolio that maximizes returns for a given level of risk. The mean-variance optimization is based on the efficient frontier, which plots the trade-off between risk (standard deviation) and expected return. The optimal mean-variance portfolio is compared to the equally weighted portfolio to evaluate the performance differences.

Efficient Frontier and Portfolio Comparison

The first plot below shows the mean-variance efficient frontier for the portfolios. The efficient frontier represents the set of portfolios that offer the highest expected return for each level of risk. The optimal mean-variance portfolio and the equally weighted portfolio are highlighted in the graph.

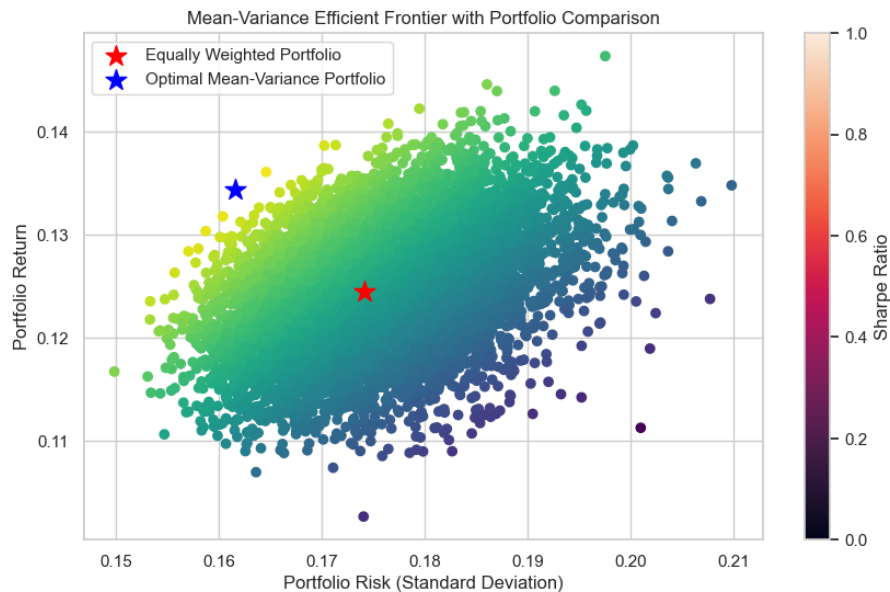


Figure 11: Mean-Variance Efficient Frontier with Portfolio Comparison

From the plot, the optimal mean-variance portfolio (marked with a blue star) lies on the efficient frontier, indicating that it provides the best return for the given risk level. The equally weighted portfolio (marked with a red star) is also plotted, and it is noticeable that it lies below the efficient frontier, implying that it is not optimized for the highest return relative to its risk.

The plot below illustrates the efficient frontier (mean-variance optimization) without the portfolio comparison. This curve represents the set of portfolios that optimally balance risk and return.

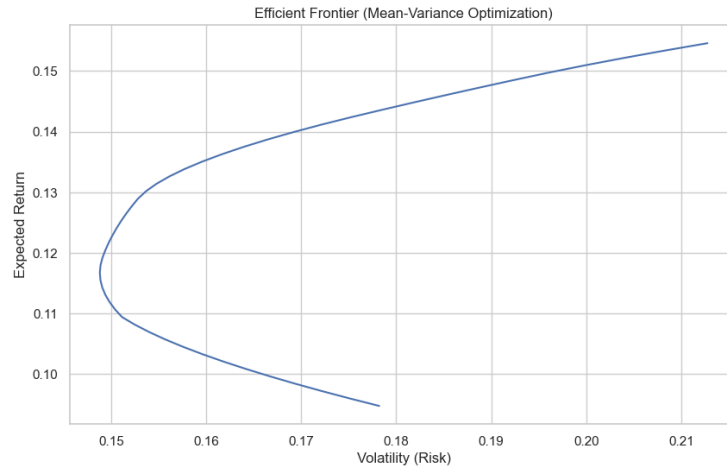


Figure 12: Efficient Frontier (Mean-Variance Optimization)

Backtesting the Optimal Portfolio

To validate the effectiveness of the optimal mean-variance portfolio, we conduct a backtest to evaluate its performance relative to the equally weighted portfolio. The backtesting results, based on historical data, show that the optimal mean-variance portfolio consistently outperforms the equally weighted portfolio in terms of risk-adjusted returns.

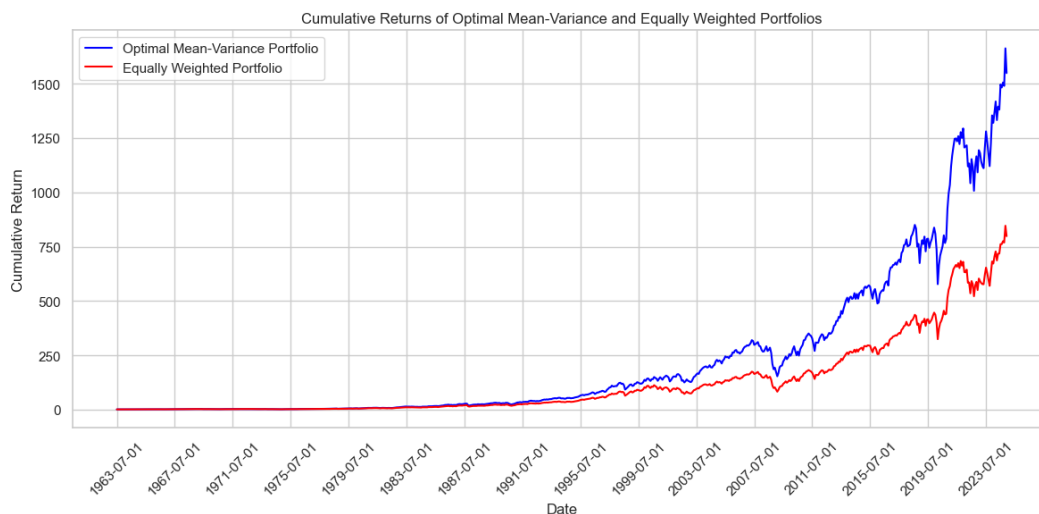


Figure 13: Portfolios Cumulative Returns Comparison

The plot clearly shows that the optimal mean-variance portfolio (in blue) significantly outperforms the equally weighted portfolio (in red) over time. The cumulative return for the optimal portfolio grows sharply after the early years, especially post-2000, reflecting the higher risk-adjusted returns achieved by the optimal portfolio.

In the earlier years (1960s to 1980s), the cumulative returns of both portfolios are relatively close, with the equally weighted portfolio showing a slight edge in the first few decades. This is because the optimal portfolio is not able to significantly outperform the equally weighted portfolio until later periods.

The performance divergence between the two portfolios becomes particularly evident after 2000, with the optimal mean-variance portfolio exhibiting exponential growth, peaking around 2020. This substantial outperformance indicates that the mean-variance optimization strategy is effective in capturing returns during the later part of the period, likely due to its dynamic adjustment to market conditions.

Interpretation:

The backtesting results show that the optimal mean-variance portfolio outperforms the equally weighted portfolio due to better risk-return trade-offs, especially in the later periods of the sample. While the equally weighted portfolio may perform similarly in stable markets, the mean-variance optimization allows the optimal portfolio to adapt to changing market conditions, capturing higher returns during periods of market growth, particularly after the year 2000.

This supports the idea that portfolio optimization using mean-variance theory can offer superior performance in terms of risk-adjusted returns, especially when the market experiences higher volatility or changing dynamics.

5 Conclusions

This research provided a comprehensive analysis of various statistical models applied to six different portfolios constructed based on size and operating profit margins. The key findings are summarized as follows:

- The returns of all six portfolios are non-normal, with distributions showing significant skewness and heavy tails. Among the models tested, the **Student's t-distribution** and **Normal Inverse Gaussian (NIG)** models best captured the return dynamics, as they are capable of accommodating mild skewness and heavy-tailed properties in the data.
- Volatility modeling was crucial, as all portfolios exhibited time-varying volatility. The **AR-GJR-GARCH with Student's t innovations** model was the best at capturing the volatility dynamics, especially during periods of market stress, suggesting that volatility clustering and asymmetric volatility effects are present in these portfolios.
- Multivariate time series analysis using the **Vector Autoregressive (VAR)** model showed that this model was appropriate for modeling the relationships between the portfolios' returns. The test for cointegration revealed that only two pairs, **ME1 OP2 vs SMALL HiOP** and **SMALL LoOP vs ME1 OP2**, showed significant cointegration at the 10% level, making them suitable candidates for statistical arbitrage strategies.
- The **pairs trading** strategy based on cointegration was found to be effective in generating high returns, despite the high risk associated with these strategies. The mean-variance optimization strategy also outperformed the equally weighted portfolio in terms of both returns and risk-adjusted performance. The strategy performed particularly well during periods of market disruption, such as the 2008 financial crisis.
- A **rolling window** approach was adopted for pairs trading, enabling the strategy to adapt to the time-varying nature of the relationships between portfolios. The strategy showed positive cumulative returns, especially during times of market volatility.

In conclusion, this research demonstrates the effectiveness of advanced time series models and optimization strategies, such as GARCH, VAR, and mean-variance optimization, in analyzing and generating profitable trading strategies for portfolios based on their size and operating profit margins.

References

- [1] Eugene F. Fama. *Efficient Capital Markets: A Review of Theory and Empirical Work*, The Journal of Finance, 25(2), 383–417, 1970. 10.2307/2325486. <https://doi.org/10.2307/2325486>
- [2] Nicolas Huck, Afawubo. *Pairs trading and selection methods: Is cointegration superior?*, Applied Economics, 47, 2015. 10.1080/00036846.2014.975417.
- [3] Robert F. Engle. *Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation*, Econometrica, 50(4), 987–1007, 1982. 10.2307/1912773. <https://doi.org/10.2307/1912773>
- [4] Tim Bollerslev. *A Conditionally Heteroskedastic Time Series Model for Speculative Prices and Rates of Return*, The Review of Economics and Statistics, 69(3), 542, 1987. 10.2307/1925546. <https://doi.org/10.2307/1925546>